

FacturadorE Backend

Two-sprint review

Sprint A — Onboarding & fiscal safety

14–18 Jul 2026

Sprint B — Recovery, seed backup & polish

19–29 Jul 2026

Presenting

Facundo Tripelhorn · Cursor Agent

Completed issues – both sprints

Sprint A · Onboarding & fiscal safety · 14–18 Jul

- [FAC-35](#) Per-profile setup state and onboarding guard
- [FAC-40](#) Production confirmation and safety cues
- [FAC-37](#) Certificate upload onboarding step
- [FAC-48](#) Block issuance when local profile is behind ARCA

Sprint B · Recovery, seed backup & polish · 19–29 Jul

- [FAC-44](#) Encrypted profile seed backup and manifest
- [FAC-45](#) S3 storage adapter for encrypted profile seeds
- [FAC-47](#) Trigger encrypted seed backup on profile config changes
- [FAC-65](#) Rebuild register from ARCA (reconstruct-based restore)
- [FAC-66](#) Portable arca-kb ARCA knowledge skill

- [FAC-38](#) Emisor and point-of-sale onboarding
- [FAC-10](#) Snapshot issuer fields on invoices at draft creation
- [FAC-52](#) Persist complete immutable invoice rendering snapshot
- [FAC-53](#) Version PDF renderers and make generated PDFs disposable

- [FAC-68](#) GET /invoices/arca read-only register peek
- [FAC-70](#) Redirect incomplete-setup HTML routes to /setup
- [FAC-71](#) Sort País destino and CUIT país dropdowns by description
- [FAC-72](#) Actionable 503 when PDF render lacks WeasyPrint/GTK
- [FAC-73](#) Default Ingresos Brutos to Exento on emisor forms
- [FAC-74](#) Fix UTF-8 mojibake in emisor templates from FAC-73

Facundo Tripelhorn

Platform · identity · recovery

Selected impact tickets from Sprints A & B

FAC-35 · Setup state & onboarding guard

Problem

A half-configured profile could still reach invoice/ARCA routes, producing confusing failures instead of a guided setup path.

Backend changes

- Added per-profile setup state (`ready` / incomplete) with a request guard.
- Incomplete HTML routes redirect to `/setup` ; JSON APIs return **503** with `setup_state` .
- `GET /health` and `GET /setup` remain available for diagnostics.

Impact

Operators cannot emit until the profile is ready. Failures become setup guidance rather than opaque ARCA/config errors.

FAC-44 + FAC-45 · Encrypted seed + S3

Problem

Profile config (emisor, PV, backup settings) lived only on disk. Rebuilding a workstation lacked a portable, encrypted config artifact.

Backend changes

- `seed.age` backup: config + manifest, age-encrypted to `recipients.txt`.
- Seed excludes DB/PDFs — ARCA remains authoritative for the register.
- S3 adapter uploads/downloads the seed without changing crypto or seed shape.

Impact

Config can travel safely between machines. Restore is a defined seed + (later) ARCA rebuild path, not an ad-hoc file copy.

FAC-65 · Rebuild register from ARCA

Problem

After seed restore, the local invoice register could be empty or stale while ARCA still held the authoritative series.

Backend changes

- Full rebuild via launcher `--restore`: identity gate → apply seed → wipe → batched `FEXGetCMP 1..N`.
- Catch-up API/UI for FAC-48 remediation (`N_local+1..N_arca`).
- Gaps (`ErrCode 1521`) recorded; transients abort after retries — never invent numbers.

Impact

A lost local DB can be reconstructed from ARCA with hard identity checks. Numbering stays ARCA-authoritative end to end.

Cursor Agent

Invoice fidelity · backup hooks · ARCA knowledge

Selected impact tickets from Sprints A & B

FAC-10 / 52 / 53 · Immutable PDF contract

Problem

Authorized PDFs could drift if live emisor/client/params changed after draft creation. Cached PDFs were also treated like durable fiscal artifacts.

Backend changes

- Snapshot issuer/display fields and `pdf_render_version` at draft creation.
- Renderers take only invoice-owned data — no live lookups.
- Versioned renderer registry; disposable local PDF cache regenerates from the snapshot.

Impact

Reprinting an authorized invoice stays faithful to what was authorized. Layout evolution is versioned; cache loss does not lose fiscal records.

FAC-47 · Seed backup on config change

Problem

FAC-44/45 could upload a seed, but nothing automatically produced a fresh remote copy when profile config changed.

Backend changes

- Event hooks on onboarding → ready, emisor edits, backup settings, `recipients.txt` replace.
- Debounced coalesce so one session of edits yields one upload.
- Persist retry state; retry on launch and via `POST /backup/seed`. Emission never triggers backup.

Impact

Remote seed stays aligned with config without operator ritual. Failures are isolated from config saves and are retryable.

FAC-66 · Portable `arca-kb` skill

Problem

Agents answered WSAA/WSFEX questions from stale training memory, inventing endpoints, ErrCodes, and TA reuse rules.

Backend changes

- Added vendor-neutral `skills/arca-kb/` with citation-backed manuals and observed homologación behavior.
- Documented methods used by FacturadorE, TA reuse, gap vs error, redacted capture shapes.
- Linked from `AGENTS.md` / repo-map so agents load it on ARCA protocol questions.

Impact

ARCA protocol answers become citation-backed and portable. Product architecture stays in `docs/design.md`; protocol trivia stops polluting agent guesses.